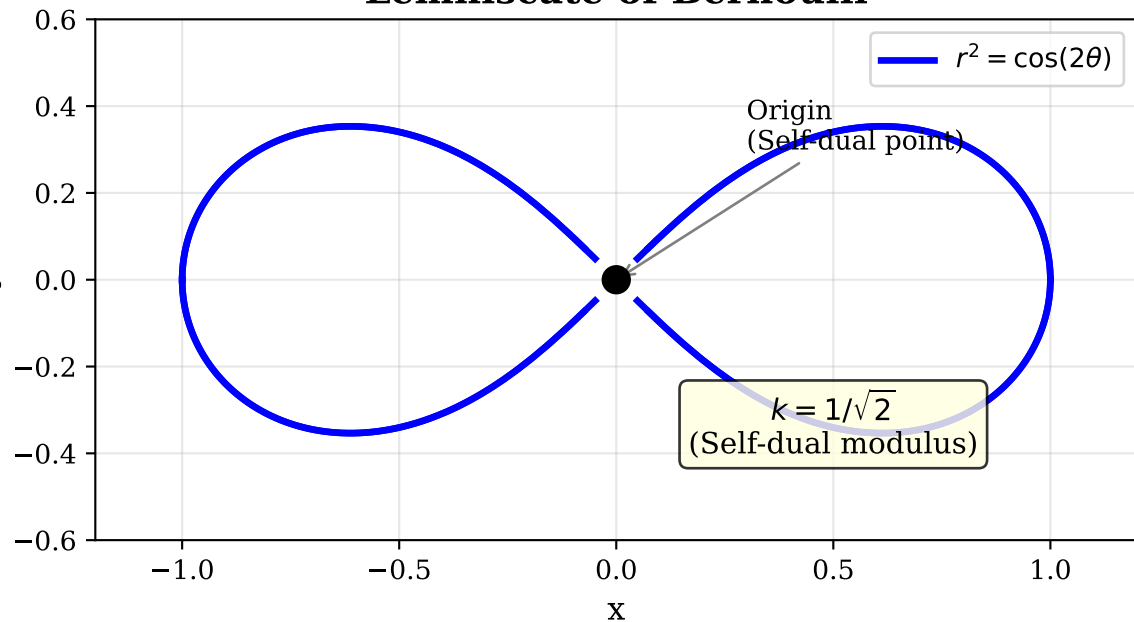


Lemniscate of Bernoulli



G* Derivation Chain

Complete Elliptic Integral $K(k)$

At self-dual: $k = 1/\sqrt{2}$

$$K(1/\sqrt{2}) = \frac{\Gamma(1/4)^2}{4\sqrt{\pi}}$$

$$G^* = \frac{\sqrt{2} \cdot \Gamma(1/4)^2}{2\pi} = 2.9587$$

Master Quadratic: $x^2 - 16G^{*2}x + 16G^{*3} = 0$

$$x_+ = 137.036 \\ = 1/\alpha$$

$$x_- = 3.024 \\ \rightarrow N_c = 3$$